

CCNA 200-301 Notes

Chapter 32 — VLAN Trunking Protocol (VTP)

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Complete chapter notes based on the supplied VTP video summary, with Cisco documentation used to verify command syntax and verification commands.

1. What is VTP?

VLAN Trunking Protocol (VTP) is a Cisco-proprietary protocol used to manage and distribute VLAN configuration information between switches in the same VTP domain.

Main idea: instead of creating the same VLAN configuration manually on every switch, VTP can distribute VLAN database changes from a VTP Server to participating switches.

VTP works across inter-switch trunk connections in the covered scenario.

2. VTP Modes

Mode	What it does	Local VLAN changes
Server	Creates, modifies, and deletes VLAN information and advertises it.	Yes
Client	Receives and stores VLAN information from the VTP domain.	No
Transparent	Does not synchronize its VLAN database with the VTP server; maintains local VLAN configuration.	Yes

3. Core Configuration Commands

Set VTP mode

```
vtp mode server
vtp mode client
vtp mode transparent
```

Set VTP domain

```
vtp domain <domain_name>
```

Set VTP password

```
vtp password <password>
```

Remove VTP password

```
no vtp password
```

Check VTP

```
show vtp status
```

4. Trunk Configuration

The inter-switch link should be configured as a trunk for the VTP lab scenario.

```
configure terminal
interface fa0/24
switchport mode trunk
```

```
end
```

Replace fa0/24 with the actual inter-switch interface.

5. Example: VTP Server

```
enable
configure terminal
vtp mode server
vtp domain CCNA-LAB
vtp password cisco123
interface fa0/24
switchport mode trunk
end
show vtp status
copy running-config startup-config
```

6. Example: VTP Client

```
enable
configure terminal
vtp mode client
vtp domain CCNA-LAB
vtp password cisco123
interface fa0/24
switchport mode trunk
end
show vtp status
copy running-config startup-config
```

7. Example: VTP Transparent

```
enable
configure terminal
vtp mode transparent
vtp domain CCNA-LAB
vtp password cisco123
interface fa0/24
switchport mode trunk
end
show vtp status
copy running-config startup-config
```

8. Verification Commands

Command	Purpose
show vtp status	Main VTP verification: mode, domain, VLAN information and Configuration Revision.
show vtp counters	Shows counters for VTP messages sent and received.
show vtp interface	Shows VTP status/configuration for interfaces.
show vtp password	Displays VTP password information according to platform/security behavior.
show interfaces trunk	Verifies the inter-switch trunk.
show vlan brief	Checks the local VLAN database and port assignments.

9. Configuration Revision Number

The Configuration Revision Number tracks changes to the VLAN configuration database. It can increase when VLAN database changes are made.

The major risk: an old/foreign switch with the same VTP domain but a higher revision number can advertise its VLAN database and potentially replace the VLAN information used

by the existing VTP domain.

Always check the revision with:

```
show vtp status
```

10. Revision-Reset Procedure

The lesson presents changing the VTP domain temporarily, checking that the revision becomes 0, then restoring the original domain.

```
show vtp status
configure terminal
vtp domain TEMP-DOMAIN
end
show vtp status
configure terminal
vtp domain ORIGINAL-DOMAIN
end
show vtp status
```

Do not connect an unknown/old switch to the production VTP domain until its VTP status and revision have been checked.

11. Important Topics □

- VTP Server: central VLAN database changes in the lesson.
- VTP Client: receives VLAN information from the domain.
- VTP Transparent: independent local VLAN database.
- VTP Domain: identifies the VTP administrative domain.
- VTP Password: must match when configured on participating devices.
- Trunk: inter-switch trunk is needed for VTP advertisements in the covered setup.
- Configuration Revision: critical when introducing old switches.
- show vtp status: the most important verification command.

12. Exam Notes □

- VTP is Cisco-proprietary.
- Remember the three modes: Server, Client, Transparent.
- Server = makes VLAN database changes.
- Client = receives VLAN information.
- Transparent = keeps local VLAN configuration independent.
- VTP domain names matter for synchronization.
- A VTP password must be consistent across participating switches if configured.
- Check Configuration Revision before connecting an old switch.
- Use show vtp status for the primary VTP verification.

13. Quick Revision

VTP → VLAN database management/distribution → Server / Client / Transparent → same VTP domain → trunk between switches → verify with show vtp status → check Configuration Revision before adding an old switch.

14. Complete Command Cheat Sheet

```
enable
configure terminal
vtp mode server
vtp mode client
vtp mode transparent
vtp domain
vtp password
no vtp password
show vtp status
show vtp counters
show vtp interface
show vtp password
interface
```

```
switchport mode trunk
show interfaces trunk
show vlan brief
end
copy running-config startup-config
```

15. Troubleshooting Checklist

- Run show vtp status on both switches.
- Check that the VTP domain is correct.
- Check the VTP mode on each switch.
- If a password is configured, check that it matches.
- Run show interfaces trunk to verify the inter-switch trunk.
- Check the Configuration Revision before adding an old switch.
- Run show vlan brief to verify the resulting VLAN database.

Scope note: This chapter follows the supplied video summary. Cisco documentation was used to verify command syntax and monitoring commands; exact VTP features can vary by Cisco platform and software version.